

```
#Loading Dataset
import pandas as pd
import matplotlib.pyplot as plt
url="https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df=pd.read_csv(url)
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques	female	35.0	1	0	113803	53.1000	C103	C

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.tail()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
#Finding Rows and Columns
df.shape
```

(891, 12)

```
#Statistical Outcome
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
#Finding Missing Values
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
#Identifying the datatypes of each columns
df.dtypes
```

	0
PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object

dtype: object

```
print(df.columns.tolist())
```

```
['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked']
```

```
#Identifying the count of Survived Passangers
# 0 means death 1 means Surviving
df['Survived'].value_counts()
```

	count
Survived	
0	549
1	342

dtype: int64

```
#Survival Rate
survival_rate = df['Survived'].mean() * 100
print(f"Survival Rate: {survival_rate:.1f}%")
```

Survival Rate: 38.4%

```
#Identifying the count of PassengerClass
df['Pclass'].value_counts()
```

count	
Pclass	
3	491
1	216
2	184

dtype: int64

```
# Handling Missing Values
df=df.drop("Cabin",axis=1)
```

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily	female	35.0	1	0	113803	53.1000	S

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
#filling the missing values of Age with median
df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
#Filling the Missing value of Embarked with Mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

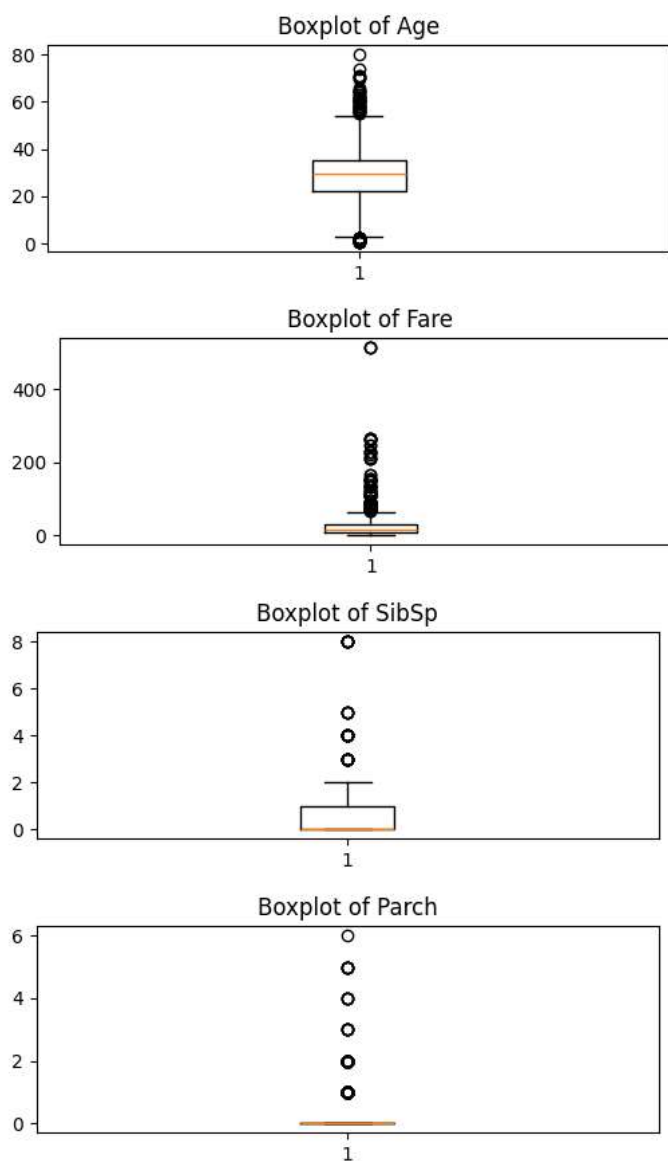
```
#After Filling Null Values Checking again if null values appears
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

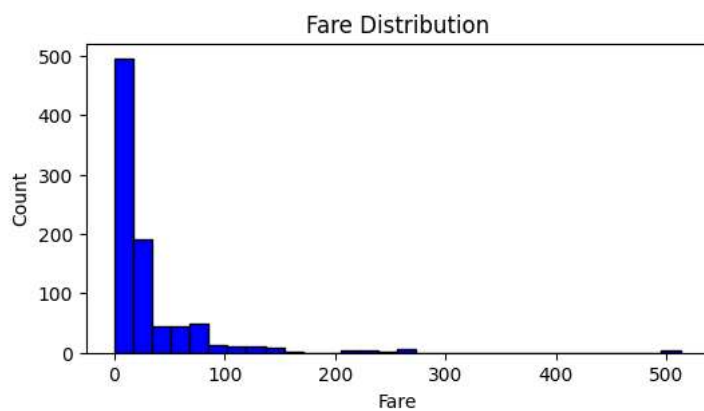
dtype: int64

```
#plotting boxplot for Numerical Columns
numeric_cols = ['Age', 'Fare', 'SibSp', 'Parch']
for col in numeric_cols:
    plt.figure(figsize=(6,2))
    plt.boxplot(df[col])
```

```
plt.title(f'Boxplot of {col}')
plt.show()
```



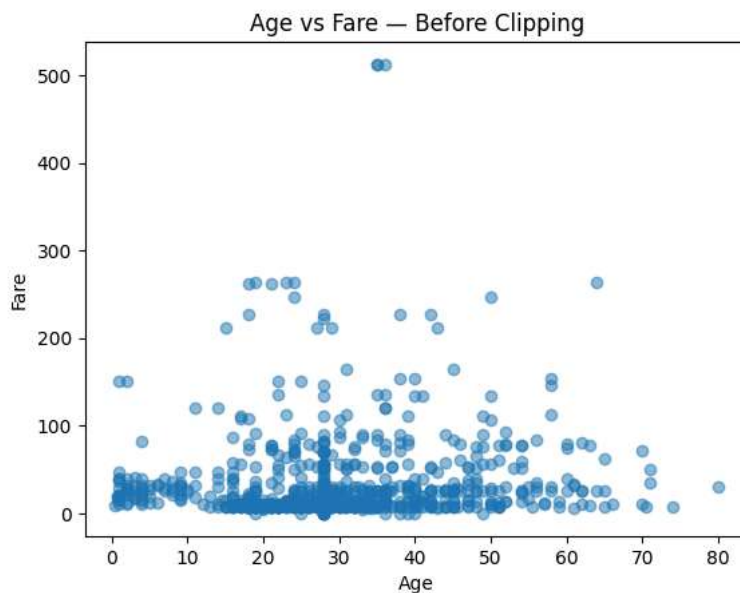
```
plt.figure(figsize=(6,3))
plt.hist(df['Fare'],bins=30,color='blue',edgecolor='black')
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```



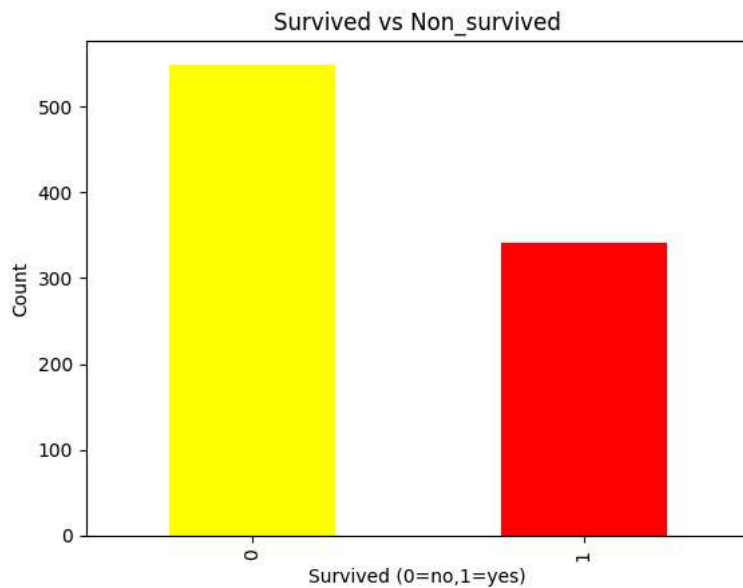
```
#Identifying the Outliers
numeric_cols = ['Age', 'Fare', 'SibSp', 'Parch']
for col in numeric_cols:
    Q1=df[col].quantile(0.25)
    Q3=df[col].quantile(0.75)
    IQR=Q3-Q1
    lower_bound=Q1-1.5*IQR
    upper_bound=Q3+1.5*IQR
    outliers=df[(df[col]<lower_bound) | (df[col]>upper_bound)]
    print(f'Number of outliers in {col}: {len(outliers)}')
```

```
Number of outliers in Age: 66
Number of outliers in Fare: 116
Number of outliers in SibSp: 46
Number of outliers in Parch: 213
```

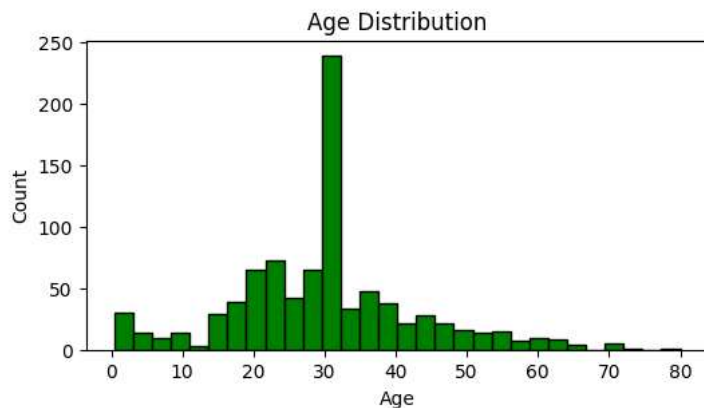
```
#Relationship between Age and fare before Clipping
plt.scatter(df['Age'], df['Fare'], alpha=0.5)
plt.title('Age vs Fare — Before Clipping')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```



```
#Plotting Barplot
df['Survived'].value_counts().plot(kind='bar',color=['yellow','Red'])
plt.title("Survived vs Non_survived")
plt.xlabel("Survived (0=no,1=yes)")
plt.ylabel("Count")
plt.show()
```



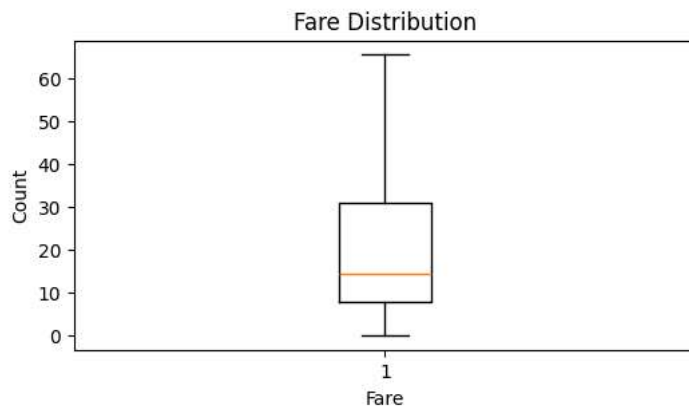
```
#plotting Histogram
plt.figure(figsize=(6,3))
plt.hist(df['Age'],bins=30,color='green',edgecolor='black')
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```



```
#clipping Outliers
Q1=df['Fare'].quantile(0.25)
Q3=df['Fare'].quantile(0.75)
IQR=Q3-Q1
lower=Q1-1.5*IQR
upper=Q3+1.5*IQR
print(lower)
print(upper)
df['Fare']=df['Fare'].clip(lower,upper)
```

```
-26.724
65.6344
```

```
#After Clipping Outliers
plt.figure(figsize=(6,3))
plt.boxplot(df['Fare'])
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Count")
plt.show()
```

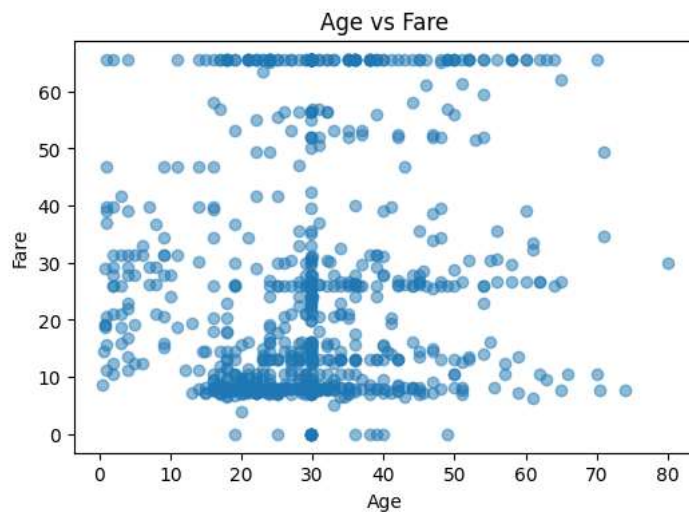


```
#Normalizing Age and Fare Column
df['Age_norm']=(df['Age']-df['Age'].min())/(df['Age'].max()-df['Age'].min())
df['Fare_norm'] = (df['Fare'] - df['Fare'].min()) / (df['Fare'].max() - df['Fare'].min())

print(df[['Age', 'Age_norm', 'Fare', 'Fare_norm']].head())
```

	Age	Age_norm	Fare	Fare_norm
0	22.0	0.271174	7.2500	0.110460
1	38.0	0.472229	65.6344	1.000000
2	26.0	0.321438	7.9250	0.120745
3	35.0	0.434531	53.1000	0.809027
4	35.0	0.434531	8.0500	0.122649

```
#Relationship between - Fare and Age after Clipping
plt.figure(figsize=(6,4))
plt.scatter(df['Age'], df['Fare'], alpha=0.5)
plt.title('Age vs Fare after Clipping')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```



```
#Conclusion
print("Shape:", df.shape)
print("Missing values:", df.isnull().sum().sum())
print("Data is clean and ready! ✅")
```

```
Shape: (891, 13)
Missing values: 0
Data is clean and ready! ✅
```

```
# =====
# CONCLUSIONS - Titanic EDA
# =====

# 1. DATASET OVERVIEW
# - 891 passengers, 12 columns
```

```
# - Mix of numerical and categorical data

# 2. MISSING VALUES FOUND & FIXED
# - Age: 177 missing → filled with median (28.0)
# - Cabin: 687 missing → dropped (77% missing!)
```